

PRINCE PATEL

Software Engineer | Backend & Distributed Systems

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Software Engineer with 1.5+ years building high-throughput distributed systems processing 125M+ records. Designs scalable microservices, ML pipelines, and cloud-native architectures on AWS. 500+ LeetCode problems solved (Top 34% globally). AWS Certified Solutions Architect.

EXPERIENCE

System Engineer — Tata Consultancy Services *Jan 2025 – Present*

- Architected an **end-to-end MLOps platform** on AWS ECS/Fargate processing 125M+ healthcare records with CI/CD via GitHub Actions, achieving zero-downtime deployments using circuit breakers and automated rollbacks
- Designed **concurrent embedding pipeline** (128 parallel workers) with SHA-256 deduplication cache and 16-shard partitioning, reducing redundant API calls by 40% and cutting Bedrock inference costs
- Built **3-tier NLP clustering system** using HNSW approximate nearest-neighbor search ($O(n \log n)$), graph-based connected components, and LLM-powered cluster labeling for semantic insight extraction
 - Engineered **auto-scaling ETL pipelines** on AWS Glue processing multi-channel data (chatbot, CRM, voicebot) with scheduled orchestration, reducing pipeline runtime by 35%
- Implemented **infrastructure-as-code** with CloudFormation across 3 environments (dev/qa/prod), managing ECS clusters, OpenSearch (3-node multi-AZ), ALB, and VPC networking
- Reduced **AWS cloud expenditure by 15%** through ECR lifecycle policies, connection pooling, batched vector upserts, and streaming I/O

Software Developer Intern — Silver Touch Technologies Ltd. *Jan 2024 – Jul 2024*

- Developed RESTful microservices using **Spring Boot** and **Python**, improving API response latency by 25%
- Built cross-platform mobile features with **Flutter** and integrated ML inference modules for real-time predictions

PROJECTS

CortexChain — LLM Application Framework | *Python, FastAPI*

- Built a **production-grade LLM orchestration framework** (120+ APIs) supporting Chains, ReAct Agents, StateGraphs with conditional routing, and multi-agent supervision
- Implemented prompt injection detection, TF-IDF retrieval, connection pooling, rate limiting, and JSONL observability

Trust Management System | *Django, PostgreSQL, Redis, WebSockets*

- Built a **full-stack financial platform** with role-based access (5 roles), multi-level approval engine, real-time WebSocket chat, and Razorpay payment integration
- Engineered TOTP-based 2FA, session management, audit logging, rate limiting (100 req/min), load-tested to p95 < 800ms

Real-Time Scalable Chat Application | *FastAPI, React, WebSockets, Tailwind CSS*

- Architected low-latency chat system using WebSocket streaming with efficient connection management and responsive UI

Plant Disease Classification Link CNN | *TensorFlow, FastAPI, JS*

- Built a CNN-based deep learning model using TensorFlow and Keras to predict plant diseases from leaf images, applying data augmentation techniques to reduce overfitting.

SKILLS

Languages:	Python, Java, JavaScript, SQL, Dart
Backend:	FastAPI, Django, Spring Boot, REST APIs, WebSockets, Microservices
Data/ML:	PySpark, Pandas, TensorFlow, NumPy, HNSW/Vector Search, NLP Embeddings
Cloud & DevOps:	AWS (ECS, ECR, Glue, Lambda, Bedrock, S3, Redshift, Aurora, CloudFormation), Docker, GitHub Actions, Auto-Scaling
Databases:	PostgreSQL, Redis, OpenSearch, ChromaDB
System Design:	Distributed Systems, ETL Pipelines, Event-Driven Architecture, Load Balancing

EDUCATION

B.E. Information Technology — Government Engineering College, Gandhinagar | CGPA: 8.65/10

ACHIEVEMENTS

- AWS Certified Solutions Architect – Associate**
- 500+ LeetCode Problems Solved** — 228 Medium, 16 Hard | Contest Rating 1,539 (Top 34%) | 28-day max streak
- 100 Days Badge 2025 & 2026** — Consistent daily problem-solving discipline